Operator: Aera Energy LLC
Well: HRT5
Type: Producer
Wellbore: 01
API : 0411104441

Wellbore Schematic

Datum: DERRICK FLOOR @ 507.80ft (default)
Ground Elev: 499.20ft
PBMD: 9ft
TD MD: 13,155ft
Orientation: Directional
Spud / KO Date: 2/9/1949

11/10/2020 ABANDONED

CASING

Size(in)	Top MD(ft)	Base MD(ft)
16.0000	9	116.0
11.7500	9	3,658.0
7.0000	3,628	12,558.0
5.0000	12,266	13,150.0

CEMENT STAGES

Job Type: Primary Date: 2/23/1949

Csg Size	Hole Size(in)	Top MD(ft)	Btm MD(ft)
5.0000	5.8750	12,266	12,709

Job Type: Squeeze Date: 3/3/1949

Csg Size	Hole Size(in)	Top MD(ft)	Btm MD(ft)
5.0000	5.8750	12,703	12,703

Job Type: Primary Date: 10/21/1947

Csg Size	Hole Size(in)	Top MD(ft)	Btm MD(ft)
16.0000		9	116

Job Type: Primary Date: 11/6/1947

Csg Size	Hole Size(in)	Top MD(ft)	Btm MD(ft)
11.7500	15.0000	1,385	3,658
11.7500	15.0000	9	697

Job Type: Primary Date: 2/3/1948

Csg Size	Hole Size(in)	Top MD(ft)	Btm MD(ft)
7.0000	10.6250	6,075	12,558

Job Type: Plug Date: 8/10/1949

Csg Size	Hole Size(in)	Top MD(ft)	Btm MD(ft)
7.0000	5.9200	11,795	12,234

Job Type: Plug Date: 10/19/2020

Csg Size	Hole Size(in)	Top MD(ft)	Btm MD(ft)
7.0000	5.9200	10,871	11,795

Job Type: Plug Date: 10/20/2020

Csg Size	Hole Size(in)	Top MD(ft)	Btm MD(ft)
7.0000	5.9200	10,215	10,871

Job Type: Plug Date: 10/26/2020

Csg Size	Hole Size(in)	Top MD(ft)	Btm MD(ft)
7.0000	5.9200	8,282	10,215

Job Type: Plug Date: 10/26/2020

Csg Size	Hole Size(in)	Top MD(ft)	Btm MD(ft)
7.0000	6.0040	7,401	8,282

Job Type: Plug Date: 10/29/2020

Csg Size	Hole Size(in)	Top MD(ft)	Btm MD(ft)
7.0000	6.0040	6,429	7,401

Job Type: Plug Date: 10/29/2020

Csg Size	Hole Size(in)	Top MD(ft)	Btm MD(ft)
7.0000	6.0940	3,644	6,429

Job Type: Plug Date: 11/9/2020

Csg Size	Hole Size(in)	Top MD(ft)	Btm MD(ft)
7.0000	6.0940	3,628	3,644

Job Type: Plug Date: 11/9/2020

Csg Size	Hole Size(in)	Top MD(ft)	Btm MD(ft)
11.7500	10.8800	2,519	3,628

Job Type: Plug Date: 11/9/2020

Csg Size	Hole Size(in)	Top MD(ft)	Btm MD(ft)
11.7500	11.0000	2,395	2,519

Job Type: Plug Date: 11/10/2020

Csg Size	Hole Size(in)	Top MD(ft)	Btm MD(ft)
11.7500	11.0000	9	2,395

PLUG DETAILS

Assembly name: PLUG

Type	Top MD(ft)	Btm MD(ft)
Cement Plug	9	2,395

Assembly name: PLUG

Type	Top MD(ft)	Btm MD(ft)
Cement Plug	2,395	2,519

Assembly name: PLUG

Type	Top MD(ft)	Btm MD(ft)
Cement Plug	2,519	3,628

Assembly name: PLUG

Type	Top MD(ft)	Btm MD(ft)
Cement Plug	3,628	3,644

Assembly name: PLUG

Type	Top MD(ft)	Btm MD(ft)
Cement Plug	3,644	6,429

Assembly name: PLUG

Type	Top MD(ft)	Btm MD(ft)
Cement Plug	6,429	7,401

Assembly name: PLUG

Type	Top MD(ft)	Btm MD(ft)
Cement Plug	7,401	8,282

Assembly name: PLUG

Type	Top MD(ft)	Btm MD(ft)
Cement Plug	8,282	10,215

Assembly name: PLUG

Type	Top MD(ft)	Btm MD(ft)
Cement Plug	10,215	10,871

Assembly name: PLUG

Type	Top MD(ft)	Btm MD(ft)
Cement Plug	10,871	11,795

Assembly name: PLUG

Type	Top MD(ft)	Btm MD(ft)
Cement Plug	11,795	12,234

UIC Project Code(s)

UIC: N/A

BASE OF USDW

Base MD (ft) N/A

BASE OF FRESH WATER

Base MD (ft) N/A

REMARKS

Primary: 5" casing cemented through port at 12709' to 12266' with 75 sks of Unknown cement at 16.2 ppg, with yield of 1.08 cf/sk. Followed by 50 sks of Colton Hi-Temp cement at 16.2 ppg, with yield of 1.08 cf/sk to Re-cement splice between 7" casing & 5" liner. Located top of cement @ 12238' & cleaned out to top of liner (12266').

Squeeze: 5" perforation squeezed at 12703' to 12703' with 50 sks of Colton Hi-Temp cement at 16.2 ppg, with yield of 1.08 cf/sk. On 3/3/49. Cmt'd through perforations in liner @ 12703'. Halliburton shoe squeeze tool @ 12670'. Displacement: holes took 4 cuft/minute at 4400#, pumped in 15 bbls of tall oil, 30 cuft of water, cmt, 10 cuft of water and 438 cuft of mud - squeezed away 40 sacks of cmt. Final pressure: 6900#. Located top of cement @ 12636'. Cleaned out cement T/ 12695', hole open T/ 12704'.

Primary: 16" Casing cemented at 116' to 8.6' with 140 sks of ictor construction cement at 15.6 ppg, with yield of 1.18 cf/sk. good cement returns to surface.

Primary: 11.75" casing cemented at 3658' to 1385' with 1000 sks of Victor construction cement at 16.2 ppg, with yield of 1.08 cf/sk. Cement returns to surface not stated. ETOC at 1385'. Primary: 11.75" casing cemented though collar at 697' to 8.6' with 500 sks of Victor construction cement at 16.2 ppg, with yield of 1.08 cf/sk. Good cement returns to surface.

Primary: 7" casing cemented at 12558' to 6045' with 1000 sks of Colton Hi-Temp cement at 16.2 ppg, with yield of 1.08 cf/sk. Cement returns to surface not stated. Adjusted cement top T/ 6045' based on CBL done on 10/23/2020.

Plug: 7" casing plugged at 12234' to 11795' with 85 sks of Hi-Temp cement through 2-1/2" tubing hung at 12,234' at Unknown ppg, with yield of 1.15 cf/sk. Ran 5-7/8' Smith bit. Located top of cement at 11,643'. Cleaned out cement to 11, 800'. Tested casing at 1000# for 15 minutes O.K. At the beginning of abandonment, TOC was tagged at 11795'.

Plug: 7" 38# casing plugged at 11795' to 10871' with 118.42 sks of Standard Blend G cement at 14.2 ppg, with yield of 1.52 cf/sk.

Plug: 7" / 38# casing plugged at 10871' to 10215' with 82.05 sks of Standard Blend G cement at 15.6 ppg, with yield of 1.56 cf/sk.

Plug: 7" / 38# casing plugged at 10215' to 8282' with 236.85 sks of Standard Blend G cement at 15.6 ppg, with yield of 1.56 cf/sk.

Plug: 7" / 35# casing plugged at 8282' to 7401' with 109.94 sks of Standard Blend G cement at 15.6 ppg, with yield of 1.56 cf/sk.

Plug: 7" / 35# casing plugged at 7401' to 6429' with 125.72 sks of Standard Blend G cement at 14.2 ppg, with yield of 1.52 cf/sk.

Plug: 7" / 32# casing plugged at 6429' to 3644' with 370.98 sks of Standard Blend G cement at 14.2 ppg, with yield of 1.52 cf/sk.

Plug: 7" casing plugged at 3644' to 3628' with .4 sks of Standard Blend G cement at 14.2 ppg, with yield of 1.52 cf/sk.

Plug: 11.75" casing plugged at 3628' to 2519' with 471.05 sks of Standard Blend G cement at 14.2 ppg, with yield of 1.52 cf/sk.

Plug: 11.75" casing plugged at 2519' to 2395' with 46.88 sks of Standard Blend G cement at 14.2 ppg, with yield of 1.52 cf/sk.

Plug: 11.75" casing plugged at 2395' to 8.6' with 1039.47 sks of Standard Blend G cement at 14.2 ppg, with yield of 1.52 cf/sk. Downsqueezed with 4.6 sks of Standard Blend G cement at 14.2 ppg, with yield of 1.52 cf/sk to build up pressure that was eventually bled to abandon.

PERFORATIONS

Top MD(ft)	Bottom MD(ft)	Density(shot/ft)
7,365.00	7,375.00	4.00
7,384.00	7,394.00	4.00
10,438.00	10,439.00	4.00
10,440.00	10,490.00	4.00
10,490.00	10,560.00	2.00
10,490.00	10,560.00	4.00
10,560.00	10,615.00	4.00
10,742.00	10,955.00	4.00
10,955.00	10,985.00	2.00
10,955.00	10,985.00	4.00
10,985.00	11,025.00	4.00
11,074.00	11,184.00	4.00
11,255.00	11,370.00	4.00
11,530.00	11,757.00	4.00
11,615.00	11,680.00	4.00
11,690.00	11,691.00	2.00
12,701.00	12,702.00	4.00
12,702.00	12,703.00	4.00
12,703.00	12,704.00	4.00
12,714.00	12,810.00	4.00
12,910.00	13,064.00	4.00

MARKER DETAILS

Name	MD top(ft)
Base of Fresh Water	227.00
A-1	1,677.00
Base of <10000ppm	1,951.00
A-2	3,823.00
A-3	5,843.00
A-4	7,020.00
D-3	7,652.57
D-4	8,140.99
T110-5	9,470.00
H14-5	10,125.00
H14-6	10,715.00
T180-6	11,255.00
T180-7	11,920.00

Casing/Slots Details	Schematic	Hole Details
CSG: 11.7500in, 54.00ppf, J-55, ST&C, 8.60ft, 34.00ft CSG: 16.0000in, 62.00ppf, H-40, , 8.60ft, 116.00ft	Ground Level	20.0000in hole to 116.00ft
CSG: 11.7500in, 47.00ppf, J-55, ST&C, 34.00ft, 2,519.00ft		
CSG: 11.7500in, 54.00ppf, J-55, ST&C, 2,519.00ft, 3,658.00ft		15.0000in hole to 3659.00ft
CSG: 7.0000in, 32.00ppf, N-80, X-LINE, 3,628.00ft, 6,437.60ft		
CSG: 7.0000in, 35.00ppf, N-80, X-LINE, 6,437.60ft, 8,290.60ft		
CSG: 7.0000in, 38.00ppf, N-80, X-LINE, 8,290.60ft, 12,365.00ft		10.6250in hole to 12365.00ft
CSG: 5.0000in, 18.00ppf, N-80, X-LINE, 12,266.00ft, 12,711.00ft		
SLOTS: 0.1500, 1.5000in, 4.0000, 12,711.00ft, 12,823.00ft		
CSG: 5.0000in, 18.00ppf, N-80, X-LINE, 12,823.00ft, 12,907.00ft SLOTS: 0.1500, 1.5000in, 4.0000, 12,907.00ft, 13,150.00ft		5.8750in hole to 13155.00ft

Obstruction	Comments	OD(in)	Top MD(ft)	Base MD(ft)
NO DAMAGE	On 11/13/12. Tight spot in casing @ 45'. Cleaned up with string mill. 7" 38# scraper and anchor still hang up slightly. (Note: 7" csg was cut & pulled F/ 3,628' during P&A on 11/4/20)	7.0000	45.00	45.00
DAMAGED CSG.	On 11/9/20. Attempted to pressure test 11-3/4" casing F/ 1,180' T/ surface. No test. Bled off pressure and chase holes. Found hole @ 335'.	11.7500	335.00	335.00
OBSTRUCTION	On 11/9/20. 11-3/4" 32-A packer and RIH F/ surface T/ 1,180' w/ (38) jts of 2-7/8" wfs. Found tight Spot @ 1,180', couldn't go thru.	11.7500	1,180.00	1,180.00
DAMAGED CSG.	On 1/3/56. Holes were found in the 7" csg @ 8,646'.	7.0000	8,646.00	8,646.00
DAMAGED CSG.	On 4/23/49. Possible bad spot in liner F/ 12,723' T/ 12,726'.	5.0000	12,723.00	12,726.00
OBSTRUCTION	On 7/12/49 – 7/14/49. Ran 3-3/4" tapered mill. Milled @ 12,904'. No progress. POOH- mill was split. Recovered small piece of liner inside mill.	5.0000	12,904.00	12,904.00
FISH	Junk left in hole: On 4/18/49. Left 2-1/2" tbg F/ 13,064' T/ 13,150'. Jarred on fish w/ 250,000#, pulled and spear came loose. POOH left (2) slips from spear in hole w/ tbg fish.	5.0000	13,064.00	13,150.00

Well Notes:

ABANDONMENT

Abandonment Tag Depth: 11,795.00ft

Initials: DM206507 Date: 12/14/2021